2.14 解答： (a) 圆柱内部区域满足拉普拉斯方程。由于势函数在原点有限，故无源圆柱内部的一般解可写为

$$\Phi(\rho,\phi)=A_0+\sum_{m=1}^{\infty}\left(\frac{\rho}{b}\right)^m\left(A_m\cos m\phi+B_m\sin m\phi\right).$$

圆柱表面的边界条件为

$$\Phi(b,\phi)=\begin{cases}V, & 0<\phi<\dfrac{\pi}{2},\\ -V, & \dfrac{\pi}{2}<\phi<\pi,\\ V, & \pi<\phi<\dfrac{3\pi}{2},\\ -V, & \dfrac{3\pi}{2}<\phi<2\pi.\end{cases}$$

该边界函数关于 ϕ 只含正弦项，因此 $A_0=0$ 且 $A_m=0$。于是

$$\Phi(b,\phi)=\sum_{m=1}^{\infty}B_m\sin m\phi,$$

其中

$$B_m=\frac{1}{\pi}\int_0^{2\pi}\Phi(b,\phi)\sin m\phi\,\mathrm{d}\phi.$$

代入分段边界条件，有

$$\begin{aligned}B_m=&\frac{V}{\pi}\left[\int_0^{\pi/2}\sin m\phi\,\mathrm{d}\phi-\int_{\pi/2}^{\pi}\sin m\phi\,\mathrm{d}\phi\right.\\ &\left.+\int_{\pi}^{3\pi/2}\sin m\phi\,\mathrm{d}\phi-\int_{3\pi/2}^{2\pi}\sin m\phi\,\mathrm{d}\phi\right]\\ =&\frac{V}{\pi m}\left[2+2(-1)^m-2\cos\frac{m\pi}{2}-2\cos\frac{3m\pi}{2}\right].\end{aligned}$$

当 m 为奇数时，$B_m=0$。当 $m=2l$ 时，

$$B_{2l}=\frac{4V}{2l\pi}\left[1-(-1)^l\right].$$

若 l 为偶数，则 $B_{2l}=0$；若 l 为奇数，令 $l=2n+1$，则 $m=4n+2$，并且

$$B_{4n+2}=\frac{4V}{\pi(2n+1)}.$$

因此边界电势可写成

$$\Phi(b,\phi)=\frac{4V}{\pi}\sum_{n=0}^{\infty}\frac{\sin[(4n+2)\phi]}{2n+1}.$$

把相应的径向因子 $(\rho/b)^m$ 乘回去，得到圆柱内部电势

$$\Phi(\rho,\phi) = \frac{4V}{\pi}\sum_{n=0}^{\infty}\left(\frac{\rho}{b}\right)^{4n+2}\frac{\sin[(4n+2)\phi]}{2n+1}.$$

(b) 为了把上式写成闭合形式，令

$$\Delta = \frac{\rho^2}{b^2}e^{2i\phi}.$$

则

$$\begin{aligned}
\sum_{n=0}^{\infty}\left(\frac{\rho}{b}\right)^{4n+2}\frac{\sin[(4n+2)\phi]}{2n+1} &= \mathrm{Im}\left[\sum_{n=0}^{\infty}\frac{\left(\frac{\rho^2}{b^2}e^{2i\phi}\right)^{2n+1}}{2n+1}\right] \\
&= \mathrm{Im}\left[\sum_{n=0}^{\infty}\frac{\Delta^{2n+1}}{2n+1}\right] \\
&= \mathrm{Im}\left[\frac{1}{2}\ln\frac{1+\Delta}{1-\Delta}\right].
\end{aligned}$$

又因为

$$\Delta = \frac{\rho^2}{b^2}\left(\cos 2\phi + i\sin 2\phi\right),$$

所以

$$\frac{1+\Delta}{1-\Delta} = \frac{1+\frac{\rho^2}{b^2}\cos 2\phi + i\frac{\rho^2}{b^2}\sin 2\phi}{1-\frac{\rho^2}{b^2}\cos 2\phi - i\frac{\rho^2}{b^2}\sin 2\phi}.$$

将分子分母同乘以

$$1-\frac{\rho^2}{b^2}\cos 2\phi + i\frac{\rho^2}{b^2}\sin 2\phi,$$

可得

$$\frac{1+\Delta}{1-\Delta} = \frac{1-\frac{\rho^4}{b^4} + i\frac{2\rho^2}{b^2}\sin 2\phi}{|1-\Delta|^2}.$$

由于分母是正实数，不改变辐角，因此

$$\arg\left(\frac{1+\Delta}{1-\Delta}\right) = \tan^{-1}\left(\frac{\frac{2\rho^2}{b^2}\sin 2\phi}{1-\frac{\rho^4}{b^4}}\right).$$

利用 $\mathrm{Im}\ln z = \arg z$，于是

$$\sum_{n=0}^{\infty}\left(\frac{\rho}{b}\right)^{4n+2}\frac{\sin[(4n+2)\phi]}{2n+1} = \frac{1}{2}\tan^{-1}\left(\frac{\dfrac{2\rho^2}{b^2}\sin 2\phi}{1-\dfrac{\rho^4}{b^4}}\right).$$

因此圆柱内部的电势也可以写成

$$\Phi(\rho,\phi) = \frac{2V}{\pi}\tan^{-1}\left(\frac{\dfrac{2\rho^2}{b^2}\sin 2\phi}{1-\dfrac{\rho^4}{b^4}}\right) = \frac{2V}{\pi}\tan^{-1}\left(\frac{2\rho^2 b^2\sin 2\phi}{b^4-\rho^4}\right).$$